

## DBS ASSIGNMENT 2

### Q1

A.

```
SELECT customer_name FROM Customers WHERE customer_id IN (  
    SELECT customer_id FROM Purchases WHERE purchase_id IN (  
        SELECT purchase_id FROM Purchase_Details WHERE book_id IN (  
            SELECT book_id FROM Books WHERE price > (SELECT AVG(price) FROM Books)  
        )  
    )  
);
```

B.

```
SELECT title FROM Books b1  
WHERE price > (SELECT AVG(price) FROM Books b2 WHERE b1.publisher = b2.publisher);
```

C.

```
SELECT customer_name FROM Customers WHERE customer_id IN (SELECT customer_id  
FROM Purchases)  
AND customer_id NOT IN (  
    SELECT p.customer_id FROM Purchases p  
    JOIN Purchase_Details pd ON p.purchase_id = pd.purchase_id  
    JOIN Books b ON pd.book_id = b.book_id WHERE b.publisher = 'Pearson'  
);
```

D.

```
SELECT title FROM Books
```

```
WHERE book_id NOT IN (SELECT book_id FROM Purchase_Details);
```

E.

```
SELECT author_name FROM Authors WHERE author_id IN (
```

```
    SELECT author_id FROM Book_Authors WHERE book_id IN (
```

```
        SELECT book_id FROM Books WHERE price > (SELECT AVG(price) FROM Books)
```

```
    )
```

```
);
```

F.

```
SELECT customer_id, SUM(quantity) FROM Purchase_Details pd
```

```
JOIN Purchases p ON pd.purchase_id = p.purchase_id
```

```
GROUP BY customer_id HAVING SUM(quantity) > 10;
```

## Q2

A.

```
SELECT p.passenger_name, b.bus_name, r.source_city, r.destination_city
```

```
FROM Tickets t JOIN Passengers p ON t.passenger_id = p.passenger_id
```

```
JOIN Buses b ON t.bus_id = b.bus_id JOIN Routes r ON t.route_id = r.route_id;
```

B.

```
SELECT p.passenger_name, t.travel_date, t.fare
FROM Tickets t JOIN Passengers p ON t.passenger_id = p.passenger_id;
```

C.

```
SELECT b.bus_name, COUNT(t.passenger_id)
FROM Buses b LEFT JOIN Tickets t ON b.bus_id = t.bus_id GROUP BY b.bus_name;
```

D.

```
SELECT r.source_city, r.destination_city, p.passenger_name
FROM Tickets t JOIN Routes r ON t.route_id = r.route_id
JOIN Passengers p ON t.passenger_id = p.passenger_id;
```

E.

```
CREATE VIEW PassengerTravelSummary AS
SELECT p.passenger_name, b.bus_name, r.source_city, r.destination_city, t.travel_date
FROM Tickets t JOIN Passengers p ON t.passenger_id = p.passenger_id
JOIN Buses b ON t.bus_id = b.bus_id JOIN Routes r ON t.route_id = r.route_id;
```

F.

```
ALTER TABLE Passengers ADD contact_number VARCHAR(15);
ALTER TABLE Tickets MODIFY fare DECIMAL(10,2);
ALTER TABLE Buses DROP COLUMN capacity;
DROP TABLE Routes;
```

**Q3**

**A. Entities & Attributes:**

- **Workout Program:** ProgramID, ProgramName, Duration, DifficultyLevel, CoordinatorName
- **Trainer:** TrainerID, TrainerName, ContactNumber
- **Fitness Center:** CenterID, CenterName

**B. Relationships:**

- **Program—Trainer:** M:N (A program has many trainers; a trainer has many programs)
- **Program—Center:** M:N (A program is at many centers; a center hosts many programs)

**C. Associative Entity (Program Trainer):**

- **TrainerAssignment:** ProgramID, TrainerID, Role

**D. ManagerName Note:**

It is stored directly in the Program table because there is a 1:1 "Program Coordinator" constraint, and coordinators are not treated as separate entities .

## Q4

**Entities:**

- **Product:** (ProductID, Name, Category, Price, Stock)
- **Outlet:** (OutletID, Name, City, Province, Manager)
- **Customer:** (CustomerID, FirstName, LastName, Street, City, Province, Zip)
- **Sale:** (SaleID, Date, Qty, Price, Total, Status, Method)

**Relationships:**

- **Outlet—Product:** M:N (Associative Entity: **Inventory**)
- **Customer—Sale:** 1:M
- **Outlet—Sale:** 1:M

## Q5

### Anomalies:

- **Insertion:** Cannot add a new Agency without an Article.
- **Update:** Updating an AI Model version requires changing many rows.
- **Deletion:** Deleting an Article might lose the only record of a Journalist.

### Functional Dependencies:

- ArticleID  $\rightarrow$  ArticleTitle, SubmissionDate
- JournalistID  $\rightarrow$  JournalistName, JournalistEmail
- AIModelID  $\rightarrow$  AIModelName, ModelVersion

### 3NF Tables:

1. **Articles:** (ArticleID, Title, Date, JournalistID, PlatformID)
2. **Journalists:** (JournalistID, Name, Email)
3. **AI\_Models:** (AIModelID, Name, Version)
4. **Agencies:** (AgencyID, Name, Phone)
5. **Verifications:** (VerificationID, ArticleID, AgencyID, Result, Score, Date)